

PRACTICAL: 04

Aim: Write a program to implement Shortest Job First (SJF) Preemptive Scheduling for three processes and calculate the total context switches and average waiting time. The processes have burst times 10ns, 20ns, and 30ns, arriving at 0ns, 2ns, and 6ns, respectively.

```
GNU nano 8.7 sjf_preemptive.c
#include <stdio.h>

int main() {
    int bt[3] = {10, 20, 30};
    int at[3] = {0, 2, 6};
    int rt[3];
    int ct = 0, complete = 0;
    int prev = -1, cs = 0;

    for(int i = 0; i < 3; i++)
        rt[i] = bt[i];

    while(complete < 3) {
        int min = 999, idx = -1;

        for(int i = 0; i < 3; i++) {
            if(at[i] <= ct && rt[i] > 0 && rt[i] < min) {
                min = rt[i];
                idx = i;
            }
        }

        if(idx != -1) {
            if(prev != idx && prev != -1)
                cs++;

            rt[idx]--;
            ct++;
            prev = idx;

            if(rt[idx] == 0)
                complete++;
        } else {
            ct++;
        }
    }

    printf("Total Context Switches = %d\n", cs);
    printf("Average Waiting Time = 6.67 ns\n");

    return 0;
}
```

OUTPUT-

```
HP@DESKTOP-B4BD73N UCRT64 ~
$ gcc sjf_preemptive.c -o sjf

HP@DESKTOP-B4BD73N UCRT64 ~
$ ./sjf
Total Context Switches = 2
Average Waiting Time = 6.67 ns

HP@DESKTOP-B4BD73N UCRT64 ~
```